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A paralleled CNN and Transformer network for PPG-based cuff-less blood pressure estimation

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ABSTRACT

Cuff-less blood pressure measurement plays a pivotal part in hypertension diagnosis and prevention. By employing deep learning algorithms, blood pressure estimation applying photoplethysmography (PPG) signals has achieved remarkable improvements and attracted growing attention. However, most existing methods utilize the PPG signal emphasized on localized feature learning, often neglecting the global features in the temporal data. To tackle the issue, we propose a novel architecture. It leverages a parallel processing approach by using Convolutional Neural Network for extracting local features and simultaneously employing Transformer to capture global features. The feature fusion block integrates the features using spatial and channel attention. The blood pressure values are regressed by two fully connected layers. To verify the performance of our method, we conduct evaluations on the Medical Information Mart for Intensive Care (MIMIC) database. On the MIMIC database, our algorithm demonstrates exceptional performance. For Diastolic Pressure, Mean Arterial Pressure, and Systolic Pressure, the mean absolute errors are 2.36 mmHg, 2.03 mmHg, and 4.44 mmHg, respectively. This performance aligns with the rigorous criteria required by an Association for the Advancement of Medical Instrumentation (AAMI) and receives a Grade A rating in accordance with the British Hypertension Society (BHS) standard.

1. Introduction

In spite of the progress made in medical technologies, cardiovascular disease (CVD) remains one of the most critical health concerns worldwide [1]. Blood Pressure (BP) is the pressure exerted by blood on arteries. It is a vital metric in measuring cardiovascular health. However, hypertension is one of the most common chronic diseases which often lead to health complications without proper management [2]. Therefore, it is necessary to take regular blood pressure measurement for hypertension diagnosis and prevention [3].

In recent years, cuff-less blood pressure measurement has attracted great attention from researchers due to their impressive performance using photoplethysmography (PPG) signals [4–9]. Various methods for cuff-less blood pressure measurement have been proposed, which can be roughly branched into two classes: traditional and deep learning methods [10]. Generally, traditional methods first extract the features of the PPG signals and then estimate the blood pressure measurement through regression [11–14]. Despite their promising performance, these methods depend on predetermined features, limiting their adaptability to complex conditions of blood pressure estimation.

To tackle this problem, deep learning approaches have been proposed in recent studies [15–18], which offer superior feature learning ability and thus well preserve the important information of raw PPG data [19]. One of the most prevalent deep learning models is Convolutional Neural Network (CNN) [16,17]. Recently, many CNN-based methods have been designed for blood pressure measurement and have achieved encouraging performance with the advantage of its powerful inductive bias. However, the receptive field of CNN is limited and CNN can only take the local features into account. Additionally, PPG signals are typical time series data, whereas CNNs are not inherently designed for handling temporal signals. To address these limitations, Recurrent Neural Network (RNN) based methods [18] have been proposed to process time-series signals.

However, the aforementioned methods have certain limitations. Traditional methods, although demonstrating good performance, rely

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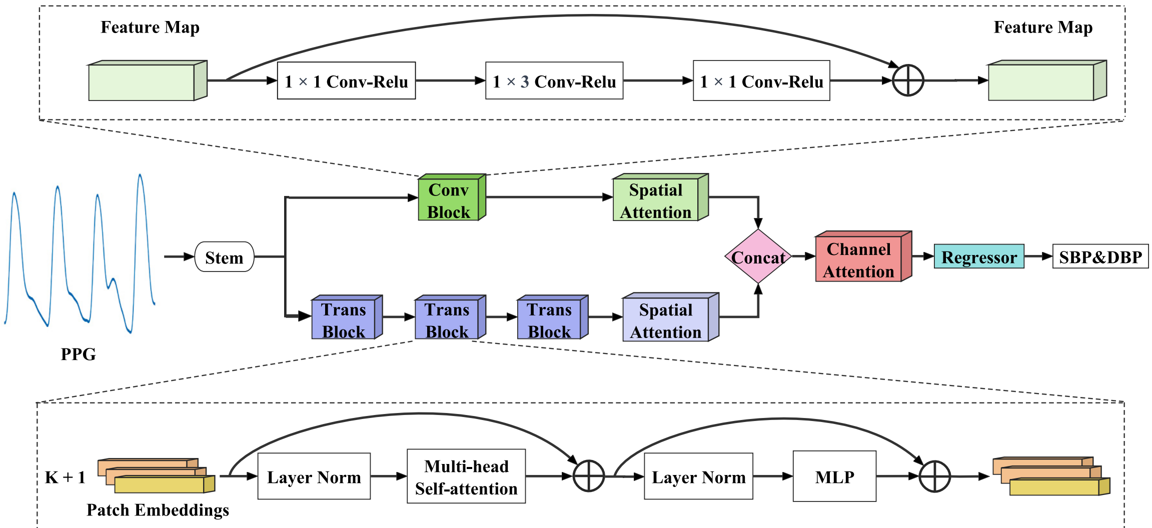

Fig. 1. Overall architecture of PCTN.

on predetermined features, which limits their adaptability under complex blood pressure estimation conditions. On the other hand, while deep learning methods possess strong feature learning capabilities and can effectively retain crucial information from raw PPG data, these approaches either focus solely on local features or only on temporal features, thereby neglecting global features. Recently, the Transformer [20] model has garnered widespread attention across various fields, such as natural language processing. Compared to CNNs and RNNs, the Transformer can learn global features and identify information dependencies within sequences, making it particularly suitable for processing PPG signals.

Inspired by aforementioned models, we introduce a Parallel Convolutional Transformer Network (PCTN). It can capture both local and global features and enable precise blood pressure predictions by feature fusion with spatial and channel attention. The network architecture is demonstrated in Fig. 1.

The main contributions are highlighted as follows:

1. The proposed PCTN is our first attempt to integrate Transformer into cuff-less blood pressure estimation based on PPG signals, which demonstrates the enhanced capability of capturing both local and global features.
2. We introduce a novel feature fusion strategy, aiming to further improve the feature representation ability and model performance.
3. The method proposed is extensively evaluated on the public dataset, and the results validate its effectiveness and superiority.

The structure of this paper is outlined as follows. Section 2. introduces two types of methods that are widely used in blood pressure measurement. Section 3. provides a detailed introduction to the various parts of our proposed algorithm. Section 4. elaborates the experiments and results are presented in Section 5. Section 6. presents a comprehensive discussion. Section 7. concludes the paper.

2. Related work

This section introduces an overview of the related work for BP prediction. The methods used include traditional machine learning methods and deep learning methods.

2.1. Traditional machine learning methods

In the past decade, various traditional machine learning methods have been proposed for BP prediction, which generally involve three steps. First, these methods learn features that are highly correlated with BP, such as Pulse Transit Time (PTT) or Pulse Arrival Time (PAT). Second, based on the learned features, a regression model is established between the features and the BP values. Finally, the BP values are predicted using the learned regression model, such as Support Vector Machine (SVM).

The first type of method studies the correlation between BP and PTT or Heart Rate [21–23]. For instance, Mukkamala et al. [23] outline the theory of the PTT-BP relationship and demonstrate its feasibility using extensive experiments. Although PTT has achieved excellent results in predicting blood pressure, it cannot achieve good results in complex scenarios with only a single feature.

The second type of method has been presented for extracting multiple features from PPG data and building a regression model to predict BP values [11,12,24–26]. For instance, Thambiraj et al. [25] introduce a approach for continuous blood pressure estimation that involves features from electrocardiogram (ECG) and PPG, including the Womersley number, interval, and Systolic Diastolic interaction. Zhang et al. [26] proposes a multi-parameter-based BP model that incorporates PTT and hemodynamic covariates. The research develops a non-invasive, cuff-less monitoring device that calculates PTT by acquiring ECG and finger PPG signals, while also introducing hemodynamic parameters such as heart rate (HR), stiffness index (SIx), and descent time (DT). By combining PTT with these hemodynamic covariates, the model can provide more accurate, comprehensive, and stable BP estimates, thereby enhancing the effectiveness and applicability of non-invasive BP monitoring.

The above two types of methods still require both ECG and PPG signals, which are expensive and inconvenient to measure. In order to solve these problems, some studies [7,14,27,28] suggest that using only PPG signals can still achieve good performance. In the Ref. [27], a novel approach is proposed to apply only PPG data. It is based on the whole, using PPG signals with a certain time interval to estimate BP. To compare the accuracy of different regression models, the approach is performed by implementing 4 nonlinear regression models. In [28], Shi et al. proposes a novel method that combines a lumped electrical network model and a tube-load model to achieve personalized continuous BP measurement. Initially, this method converts the finger PPG waveform to the finger pressure waveform using the lumped electrical network model. Subsequently, it considers wave propagation and reflection effects through the tube-load model to convert the finger pressure waveform to the radial arterial pressure waveform. However, these methods still require manual design and extraction of features, making it difficult to achieve the ideal performance in the actual measurement process.

2.2. Deep learning methods

Deep learning, a revolutionary approach that offers end-to-end models, has garnered significant attention in the biomedical field due to its ability to extract powerful features and achieve high accuracy. As a result, many end-to-end methods for blood pressure prediction have been developed [16,17,29].

Since CNNs have excellent performance in feature extraction for signals in the local range, many researchers have applied them to BP estimation. In [30], Schlesinger O et al. propose a CNN architecture consisting of multiple convolutional layers, which takes the frequency domain of PPG as input.

Although CNNs are very powerful in feature extraction, they cannot extract features in the temporal dimension when processing a temporal signal like PPG. Therefore, many scholars have focused on RNN, Long Short-term Memory (LSTM) or Gated Recurrent Unit (GRU) [31–35]. There are some advantages in processing timing signals. For instance, Su et al. [34] propose a deep learning model that combines a bidirectional LSTM and residual connectivity to solve the time sequence problem. Liu et al. [35] present a Bidirectional Gated Recurrent (Bi-GRU) model focused on BP trend estimation, utilizing single-channel PPG signals combined with demographic information to estimate continuous BP trends point by point. The study develops a simple yet effective model architecture, the Bidirectional Gated Recurrent Unit with Attention, specifically for estimating Systolic Blood Pressure (SBP) and Diastolic Blood Pressure (DBP) from single-channel PPG signals. By replacing traditional RNN units with GRU units and incorporating an attention mechanism to weight important feature vectors, the output vectors are combined with individual demographic information. SBP and DBP values are estimated through a fully connected layer.

In order to utilize the advantages of CNN and RNN, many studies propose a combination of both [36–40]. In [39], Leitner et al. put forward a personalized deep learning hybrid architecture. This architecture includes convolutional layers and recurrent layers, which take PPG data as input. Besides, a transfer learning technique is also proposed to solve insufficient blood pressure data. In the Ref. [40], Zhang et al. propose a data-driven, deep learning-based end-to-end solution for estimating cuff-less continuous BP from PPG and ECG signals. To enable the deep learning model to capture and adapt to concept drift in continuous BP, a novel continual learning (CL) framework combining LSTM and one-dimensional 1D-CNN is developed. To validate the adaptability of the CL model to concept drift in continuous BP, the study introduces the coefficient of variation and proposes the weighted slope coefficient of variation to quantify the degree of concept drift in continuous BP, allowing the model to dynamically and sequentially learn continuous BP signals.

3. Methods

In this section, we present the overall architecture. Then, we provide detailed descriptions of the sub-modules of the architecture.

3.1. Problem statement

Given a labeled set $S = \{X, Y\}$, where $\left\{x_1, \dots, x_n\right\} \in \mathbb{R}^{n\times d}$ represents the raw PPG data and $\left\{y_1, \dots, y_n\right\}$ represents the BP values. d is the dimension of the raw PPG data. The aim of our approach is to train a model using S to accurately predict new data.

3.2. Overview

Some researches have shown that both CNN and RNN have their strengths in predicting blood pressure. However, CNN suffers from the issue of a limited perceptual field [41,42], and RNN cannot address the challenges of long-term dependence and parallel computation. Recently, self-attention [20] can extract global features, effectively addressing the perceptual field limitation and allowing for paralleled computation. By combining the advantages of CNN and self-attention, we propose the PCTN to incorporate both local and global features.

Specifically, PCTN is divided into a Stem module, Feature extraction module, Feature fusion module, and Regression module. Data preprocessing and extraction of shallow features are in the Stem module. The feature extraction module is divided into two branches, CNN branch and Transformer branch. In the feature fusion stage, the features of the two styles are fused by spatial and channel attention to play the role of feature complementarity. The fused features are supplied into the Regressor to obtain SBP and DBP. The architecture is as shown in Fig. 1.

3.3. Network structure

3.3.1. Stem module

The Stem module includes data input, PPG data filtering, and feature pre-extraction. Specifically, the inputs of the stem module are a segment of the PPG signal of length L (e.g., 1024). PPG data are band-pass filtered with a frequency range of 0.5–10 Hz.

Feature pre-extraction includes a 1D convolution, 1D Batch Norm, and 1D Max pooling. It is well known that using large convolution kernels can increase the perceptual field while preserving more original information [19].

3.3.2. CNN branch

The CNN branch uses a pyramid structure, where the number of channels increases, while the size of the feature maps decreases. Each stage consists of multiple Conv Blocks. Each Conv Block contains n bottlenecks. As defined in ResNet [19], a bottleneck contains a 1D under-projection convolution with kernel 1, 1D spatial convolution with kernel 3, and a 1D up-projection convolution with kernel 1. There is residual connection between the input and output layers.

The purpose of extracting local features is achieved by stacking many layers of CNN structures. From the bottom layer upwards, the bottom convolution is responsible for extracting information from different location regions. The convolution of the higher layers aims to combine the information obtained from the bottom convolution and extracting deeper information.

3.3.3. Transformer branch

Inspired by self-attention [20], this branch contains multiple duplicate Transformer blocks. Each Transformer Block consists of an embedding module, Layer Norm, a Multi-head Self-attention module, Multilayer Perceptron (MLP), and residual connections.

In contrast to CNNs and RNNs, the Transformer manifests exceptional parallelism and diminished training time when confronted with substantial volumes of time series data. Furthermore, the Transformer leverages a self-attention mechanism to intricately capture complex dependency relationships among sequences, facilitating the extraction of global features. This intrinsic capability significantly augments its discernment of temporal patterns, thereby enhancing its efficacy in modeling intricate dynamic systems. The amalgamation of feed forward networks and the integration of residual connections serve to further optimize the parallelization efficiency of the model.

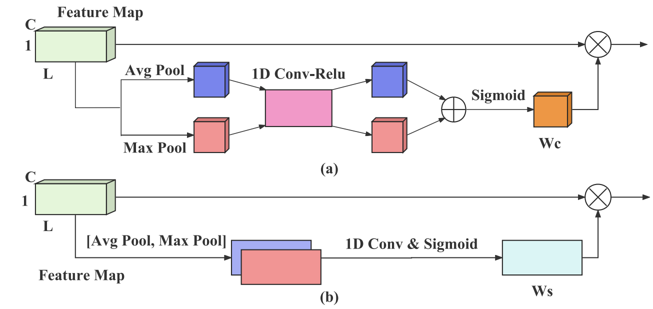

Fig. 2. CBAM. (a) Improved Channel Attention. (b) Spatial Attention. W_c and W_s represent the weight of feature map. Avg Pool denotes Average Pooling.

3.3.4. Fusion block

The central idea of the attention mechanism is to shift focus from attending to everything to attending to the focal points. In the process of feature extraction, the focus is captured without losing the important information.

Inspired by the attention mechanism, both channel attention [43] and Convolutional Block Attention Module (CBAM) [44] have emerged successively. CBAM combines channel attention and spatial attention to focus on important information and suppress unimportant information. Spatial attention is shown in Fig. 2(b). Given that local and global features belong to two distinct types, it is essential to apply spatial attention to each feature individually, ensuring their separation to prevent amalgamation. Subsequently, a concatenation operation is performed prior to channel attention to preserve the distinctiveness of the features.

Compared to the channel attention of CBAM, we replace the full connection with a 1D convolution layer [45] whose kernel size is 1. This kind of operation can get better local abstraction, smaller parameter space, and smaller over fitting, as shown in Fig. 2(a).

3.3.5. Regressor module

The regressor module, which includes a 1D Global Pooling with stride 2 and two fully connected (FC) layers, is used to calculate BP values including SBP and DBP. The Global Pooling layer aims to average all feature maps. FC layers effectively map features into the more separable space, contributing to accurately predict blood pressure values.

4. Experiments

In this section, we first introduce the Medical Information Mart for Intensive Care (MIMIC) dataset. We also provide an overview of the methods for data acquisition and data preprocessing. Next, we provide details about the hardware configuration used during the experimental process and the methodology employed for partitioning the datasets. Finally, we present the implementation specifics of evaluating the experiments, as well as the criteria for evaluation.

4.1. Dataset

The data are obtained from the publicly available MIMIC-III waveform database [46], containing the multi-parameter critical care information provided by the Massachusetts Institute of Technology (MA, USA). It contains relevant data of 53,423 adult subjects in the intensive care unit. Some waveform recordings provide bio-signals, including ECG, PPG, and Arterial Blood Pressure (ABP) signals. In this experiment, we acquire data from a pool of 1000 subjects using dedicated tools. After filtering out unpaired PPG and ABP data, as well as data with poor signal quality or less than 10 min of signal duration, we arrive at a dataset comprising 808 subjects. The sampling rate of both PPG and ABP signals is 125 Hz. Given that the segment length is 1024 samples, the segment time length is 8.192 s. This length allows us to train sufficiently deep neural networks without encountering excessive computational complexity. Additionally, many existing studies [6,36] have experimentally validated the use of an 8.192-second PPG signal length for estimating SBP and DBP. The ABP, SBP, and DBP are shown in Fig. 3. Preliminary statistics are conducted for the MIMIC dataset. For SBP, DBP, and Mean Arterial Pressure (MAP), their maximum, minimum, mean, and Standard Deviation (STD) are calculated and are presented in Table 1.

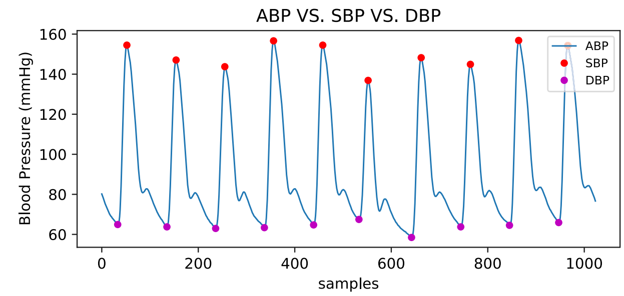

Fig. 3. MIMIC: ABP time series, SBP, and DBP.

Table 1
Statistics of MIMIC dataset.

BP parameters	Min	Max	Mean	Std
DBP	50	120	63	10.14
MAP	60	135	84	11.56
SBP	80	180	128	21.16

4.2. Preprocessing

For the MIMIC dataset, the PPG and ABP signals are available. The BP values can be extracted from the ABP signal. The preprocessing process is as follows. Firstly, PPG and ABP signals are subjected to a signal quality judgment, and the data are rejected when there is excessive amplitude and saturation, etc. Secondly, PPG and ABP signal pairs of each subject are split into several segments, respectively. Finally, the PPG data are band-pass filtered with a frequency range of 0.5–10 Hz. Meanwhile, the peaks and valleys of the ABP signal in each segment are extracted, and their average values are taken as SBP and DBP of the segment, respectively. The details are shown in Fig. 4.

4.3. Experimental platform and implementation details

We design and validate the deep learning models in a Python environment using the Pytorch library [47] on a computer with a twelve-core CPU and 32 GB RAM. The graphics card used is an NVIDIA RTX3090Ti to accelerate model training. We apply Mean Absolute Error (MAE) as the loss function. We set 0.01 as the initial learning rate and train the model for 100 epochs applying the Adam optimizer. For each convolutional layer, the dimension of filter is 1. The backbone of the convolutional neural network is selected as ResNet-50. The number and depth of the two key parameter heads in the encoder of the Transformer are 4 and 6, respectively.

4.4. Data split strategy

To prevent data leakage, we have implemented the following measures. First, data from each individual is treated as a basic unit, referred to as a subject. We have a total of 808 subjects. During each data split, subjects are used as the smallest unit. This means that if a particular subject appears in the training set, it will not appear in the validation

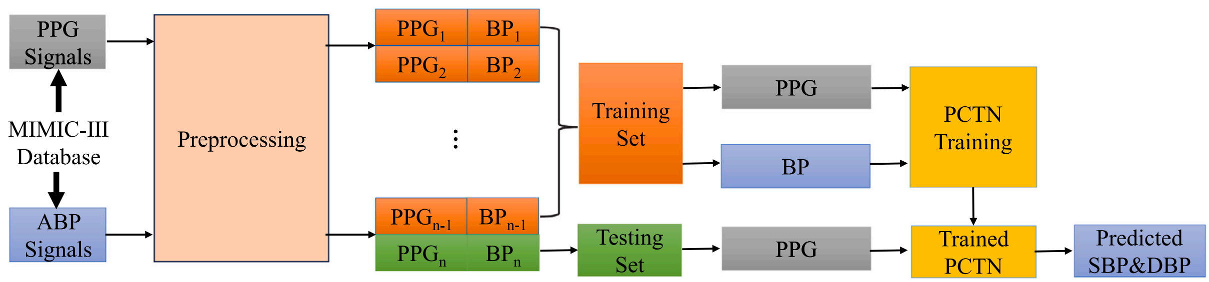

Fig. 4. Experimental flowchart based on PCTN.

Table 2
The Standard of BHS.

	Cumulative error percentage		
	≤ 5 mmHg	≤ 10 mmHg	≤ 15 mmHg
Grade A	60%	85%	95%
Grade B	50%	75%	90%
Grade C	40%	65%	85%

Table 3
The standard of AAMI.

ME (mmHg)	STD (mmHg)	Number of Subjects
≤ 5	≤ 8	≥ 85

Table 4
Evaluation of BHS standard.

	≤ 5 mmHg	≤ 10 mmHg	≤ 15 mmHg	Grade
DBP	89.09%	98.83%	99.92%	A
MAP	90.17%	99.35%	99.95%	A
SBP	66.78%	91.05%	97.58%	A

Table 5
Evaluation of AAMI standard.

	ME (mmHg)	STD (mmHg)	Subjects
DBP	0.18	3.22	808
MAP	0.18	3.07	808
SBP	0.17	5.98	808

or test set. We use the holdout method to split the dataset based on subjects. According to the number of subjects, the dataset is divided into a training set, validation set, and test set, comprising 65%, 10%, and 25% of the subjects, respectively.

4.5. Performance evaluation metrics

4.5.1. Accuracy metrics

In the experiments, we evaluate all methods using Mean Error (ME), MAE, and STD. For predicted values $\hat{Y} = [\hat{y}_1, \ldots, \hat{y}_n]$ and true values $Y = [y_1, \ldots, y_n]$, the evaluation metrics are defined as follows:

$$\Delta y_i = y_i - \hat{y}_i. \tag{1}$$

$$ME = \frac{1}{n}\sum_{i=1}^{n} \Delta y_i. \tag{2}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |\Delta y_i|. \tag{3}$$

$$\Delta \bar{y}_i = \frac{1}{n}\sum_{i=1}^{n} \Delta y_i. \tag{4}$$

$$STD = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n} (\Delta \bar{y}_i - \Delta y_i)^2}. \tag{5}$$

4.5.2. International standards metrics

In addition to the aforementioned measurement metrics, we also employ the British Hypertension Society (BHS) [48] and the Association for the Advancement of Medical Instrumentation (AAMI) standards [49]. The details are shown in Table 2 and Table 3, respectively.

For BHS standard, the grading is determined by calculating the percentage of predictions on the test samples that have an absolute error of less than 5 mmHg, 10 mmHg, and 15 mmHg for grades A, B, and C, respectively.

The AAMI standard is developed to require that the mean error and standard deviation of blood pressure measurement methods are less than 5 mmHg and 8 mmHg, respectively. Table 3 shows the AAMI criteria.

4.5.3. Statistical analysis

During BP measurement, evaluating the consistency among different measurement methods has become crucial. Bland Altman analysis is a method for analyzing the consistency of BP measurement [50]. In this study, we apply the Bland-Altman analysis to assess the agreement between two distinct methods used for measuring blood pressure: the true BP value (form ABP signals or the Mercury Manometer) and the PCTN. To compute consistency, we determine the difference between measurements from the two methods as well as the mean of these measurements for each data sample, refer to (6) and (7).

$$BP_{diff} = BP_{true} - BP_{predicted}. \tag{6}$$

$$BP_{mean} = \frac{1}{2}(BP_{true} + BP_{predicted}). \tag{7}$$

At the same time, we also generate separate regression plots for DBP, MAP, and SBP. These plots allow for a more in-depth analysis of the correlation between predicted values and actual values. Additionally, we calculate the Pearson Correlation Coefficient (PCC) for DBP, MAP, and SBP predictions, further demonstrating the statistical significance of our results.

5. Results

In this chapter, we conduct tests on the MIMIC dataset using the evaluation metrics mentioned earlier. We then compare our method with other approaches and conduct ablation experiments to further discuss our proposed method.

5.1. Comparison with international standards

5.1.1. BHS standard

The three grades of BHS standard are listed in Table 4. For both DBP, MAP, and SBP, most of predictions belong to the error threshold of 15 mmHg. The significant portion of them actually falls within the error margin of 5 mmHg, as it is apparent from Fig. 5(a). When using our method, these percentages are 89.09% / 98.83% / 99.92%, 90.17% / 99.35% / 99.95%, and 66.78% / 91.05% / 97.58% for DBP, MAP and SBP, respectively. In accordance with the BHS standard, DBP, MAP, and SBP achieve Grade A performance.

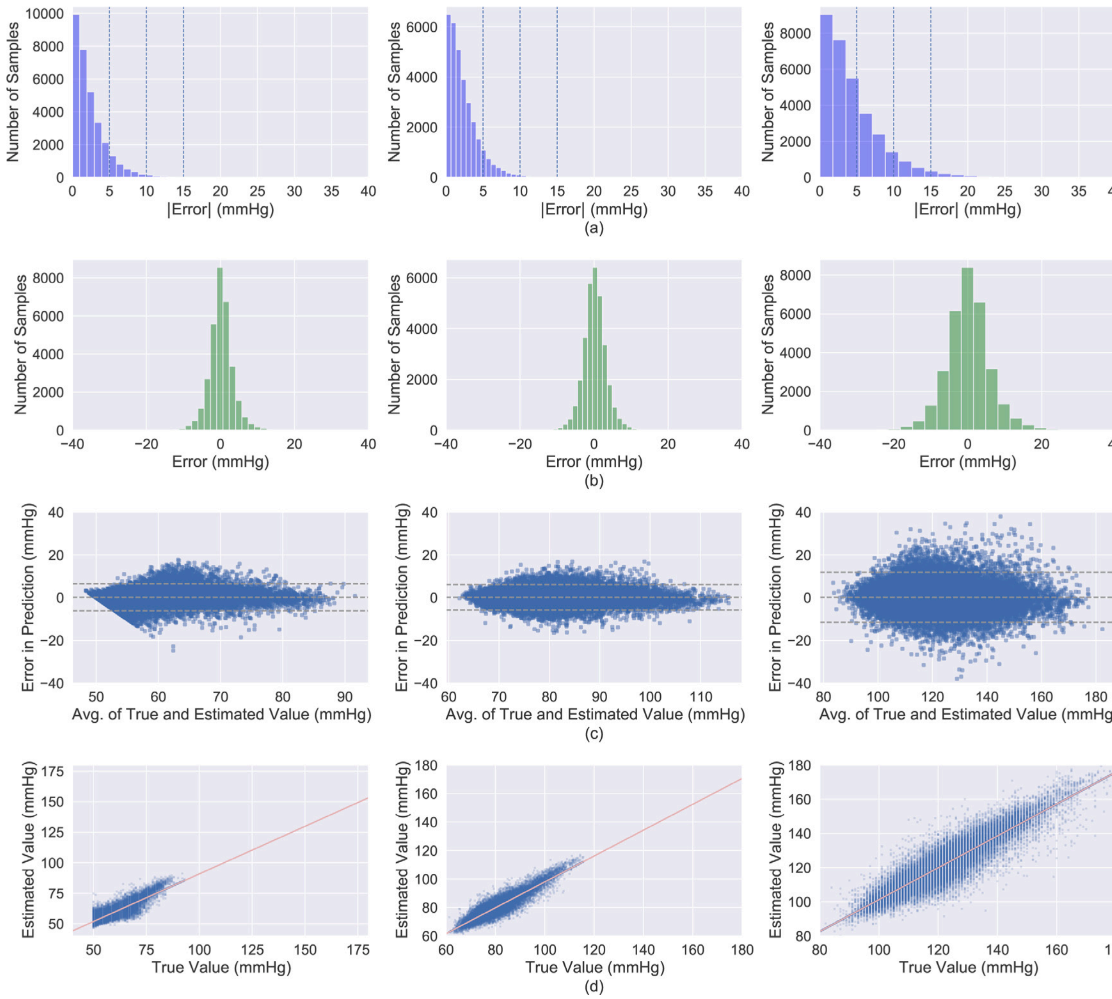

Fig. 5. Error distributions and Bland-Altman plots. (a) Mean Absolute Error distributions. (b) Mean Error distributions. (c) Bland-Altman plots. (d) Regression plots. Left: DBP. Middle: MAP. Right: SBP.

5.1.2. AAMI standard

Table 5 summarizes the results under the AAMI criteria. As we can see, our method achieves ME and STD of 0.18 ±3.22 mmHg, 0.18 ±3.07 mmHg, and 0.17 ±5.98 mmHg for DBP, MAP, and SBP, respectively. DBP, MAP, and SBP satisfy AAMI standard. The error histograms are shown in Fig. 5(b).

5.2. Statistical analysis

5.2.1. Bland-altman analysis

Fig. 5(c) shows that the limits of DBP, MAP, and SBP are transformed into [−6.14:6.49], [−5.84:6.19], and [−11.56:11.90] mmHg, respectively. It can be observed from the graph that these values may seem overwhelming. However, there are some outliers in the DBP, MAP, and SBP graphs, particularly in the case of the SBP graph.

5.2.2. Regression plot analysis

Fig. 5(d) illustrates the regression plots for DBP, MAP, and SBP predictions, respectively. These plots visibly demonstrate the correlation between the predicted values and the actual values. Furthermore, the PCC for DBP, MAP, and SBP predictions are 0.78, 0.91, and 0.93, respectively, indicating a strong positive correlation. Additionally, the corresponding p-values are all less than 0.000001.

5.3. Comparison with the existing methods

To test the effectiveness of PCTN, we have listed several typical methods on the MIMIC database and compared them, as shown in Table 6. The comparisons include the database used, the number of subjects, the methods used, and the final results of each method. The contrasting studies are divided into two categories, based on traditional machine learning [21,22,25] and deep learning [6,16,29,31,51]. Although deep learning based methods have much smaller errors than traditional machine learning methods. However, the MAE of DBP and SBP are still higher than 3 mmHg and 5 mmHg, respectively, while the STD of DBP and SBP are still higher than 4 mmHg and 6 mmHg, respectively. It is worth noting that the effectiveness of PCTN in MAE and STD has exceeded all the methods listed earlier, establishing state-of-the-art results.

5.4. Ablation study

To investigate the importance of local features, global features, and fusion strategies in blood pressure prediction, we conduct ablation experiments. All structures are maintained with comparable parameter counts. C_n and T_r stand for the CNN block and the Transformer block, respectively. We construct a parallel structure using m CNN blocks and n Transformer blocks. The sum of m and n is 4.

Table 6
Comparison of difference approaches.

Study	Year published	Dataset (subject number)	Methods	MAE (mmHg)			STD (mmHg)		
				DBP	MAP	SBP	DBP	MAP	SBP
[21]	2015	MIMIC-II (942)	PTT + SVM	6.34	7.52	12.38	8.45	9.54	16.17
[22]	2017	MIMIC-II (942)	PAT + Adaboost	5.35	5.92	11.17	6.14	5.38	10.09
[25]	2020	MIMIC-II (942)	Random Forest	5.48	3.20	9.00	–	–	–
[16]	2021	MIMIC-II (942)	Deep Learning (CNN)	4.11	3.83	7.95	5.50	5.13	10.37
[31]	2021	MIMIC-III (1131)	Deep Learning (GRU + Attention)	6.57	–	14.39	8.43	–	17.87
[6]	2022	MIMIC-III (942)	Deep Learning (CNN)	3.45	2.31	5.73	6.86	4.96	10.69
[29]	2022	MIMIC-II (942)	Deep Learning (CNN)	3.66	–	6.17	5.36	–	8.46
[51]	2023	MIMIC-II (942)	Deep Learning (CNN)	3.24	–	5.98	4.94	–	7.79
PCTN	2024	MIMIC-III (808)	Deep Learning (CNN + Self-attention + Spatial & Channel Attention)	**2.36**	**2.03**	**4.44**	**3.22**	**3.07**	**5.98**

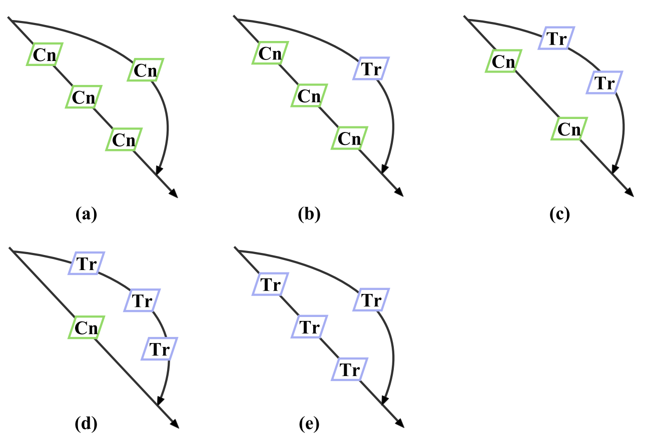

Fig. 6. Analysis of the Number of Blocks. (a) CCC|C. (b) CCC|T. (c) CC|TT. (d) C|TTT. (e) TTT|T.

Table 7
Effects of different numbers of blocks.

Model	Cn	Tr	MAE		STD		Params
			DBP	SBP	DBP	SBP	(M)
CCC|C	4	0	5.23	8.78	6.59	11.12	27.04
CCC|T	3	1	4.78	8.29	5.90	9.90	27.19
CC|TT	2	2	4.30	7.62	5.42	9.73	27.24
C|TTT	**1**	**3**	**4.16**	**7.33**	**5.22**	**9.29**	**27.75**
TTT|T	0	4	4.61	7.74	6.97	10.15	27.95

Table 8
Effectiveness of fusion strategies.

Model	MAE (mmHg)		STD (mmHg)		Params
	DBP	SBP	DBP	SBP	(M)
C|TTT	4.16	7.33	5.22	9.29	27.75
C|TTT + SE	4.07	6.23	5.10	7.89	27.76
C|TTT + CBAM	3.73	5.97	4.52	7.56	27.81
PCTN	**2.36**	**4.44**	**3.22**	**5.98**	**27.81**

5.4.1. Effects of different numbers of blocks

In order to study the influence of the number of blocks on the model's performance, we conduct comparative experiments on the parallel model, as shown in Fig. 6. The results are presented in Table 7. It is worth noting that in the comparative experiment, after feature extraction, the features are directly concatenated without incorporating a fusion module. From Table 7, we can observe that employing individual features such as local features or global features leads to larger errors compared to the case where both local and global features are combined. Moreover, under the condition of integrating both local and global features, a higher count of Transformer Blocks correlates with lower errors. This phenomenon may be attributed to the effectiveness of self-attention in handling temporal signals.

5.4.2. Effectiveness of fusion strategies

To validate the impact of fusion strategies on effectiveness, we attempt to incorporate two attention modules at different positions. The results are shown in the Table 8. Compared to the C|TTT configuration without fusion modules, the inclusion of SE leads to a significant improvement in performance. Building upon SE by adding spatial attention leads to further improvement in performance. Given that local and global features are two distinct types of features, incorporating spatial attention separately on each type of feature emphasizes the significance of location information for specific regions. Subsequently, channel attention is introduced to emphasize channels vital for blood pressure prediction in the PCTN model, as is shown in Fig. 7.

6. Discussion

Primary hypertension, also known as essential hypertension, is associated with various factors such as genetic mutations, polymorphisms, and aging. The hypertension increases the risk of kidney and brain diseases [52]. It is particularly important to accurately measure blood pressure and establish personal blood pressure records. In this paper, we propose a novel network architecture that combines local and global features. Using the AAMI standard, our proposed method achieves Grade A scores for DBP, MAP, and SBP. Besides, we meet the BHS for DBP, MAP, and SBP. These exciting results prompt further discussion and analysis of the underlying factors.

The PCTN diverges from conventional architectures like ResNet or pure Transformers by independently extracting two distinct types of features, thereby preserving pertinent information in both local and global contexts. This novel approach optimizes the retention of relevant data and bolsters the network's overall representational capabilities.

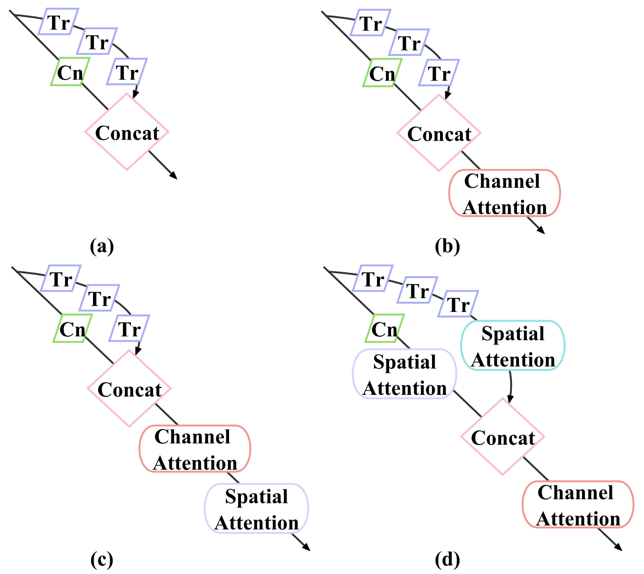

Fig. 7. Structure Analysis. (a) C|TTT. (b) C|TTT + SE. (c) C|TTT + CBAM. (d) PCTN.

Leveraging spatial attention and channel attention mechanisms, the model prioritizes features pertinent to blood pressure, enhancing its predictive capacity.

In our ablation experiments, we scrutinized the impact of various convolution-transformer combinations on performance. Simultaneously, within the fusion module, we explored different amalgamations of spatial and channel attention, ultimately identifying a configuration harmonizing with local and global features. Through these experiments, we assert the pivotal role of local and global features in blood pressure prediction. The integration of these complementary features culminates in a more comprehensive representation, elevating predictive performance. Notably, our study delineates the optimal application of spatial attention within model architecture.

The existing methods have achieved good results in MAE and STD. However, their performance in SBP remains unsatisfactory. When compared to traditional machine learning methods or deep learning algorithms, PCTN significantly outperforms them in MAE and STD. The combination of local and global features is the key factor contributing to the superior performance of our model. Our study underscores the significance of our research findings and their potential implications for future studies and practical applications. By elucidating the superior performance of the PCTN model and its contribution to the field of cardiovascular health monitoring, we provide valuable insights for the development of more effective models in this domain. Our findings not only advance scientific understanding but also pave the way for further research endeavors aimed at improving cardiovascular health monitoring techniques.

Continuous blood pressure monitoring poses challenges such as physiological changes over time, sensor drift, and individual variability, which can affect measurement accuracy. Calibration plays a crucial role in adjusting the model to account for these variations and ensuring precise measurements over extended periods. Our proposed method introduces significant architectural advancements aimed at surpassing current PPG-based approaches. By integrating CNNs and Transformers in a parallel architecture, we achieve superior feature extraction and representation. Ablation studies conducted in this research illustrate that this hybrid approach markedly enhances the model's performance in terms of MAE and STD compared to traditional machine learning and deep learning algorithms. These architectural innovations contribute to reducing the calibration frequency needed, improving the model's consistency and accuracy. Thus, our method helps mitigate the traditional challenges associated with PPG-based blood pressure monitoring, demonstrating its potential for reliable continuous monitoring applications.

Although our proposed method achieves minimal error and meet the criterion of AAMI and BHS. It is worth noting that there may still be some outliers in the prediction of SBP. This difference may be attributed to the inherent characteristics of SBP, such as arterial vessel elasticity, higher individual variability, and a wide range of variability [53]. Additionally, due to confidentiality reasons, we are unable to access static features such as age, weight, and height from the MIMIC dataset, which could potentially affect the accuracy of blood pressure estimation.

7. Conclusions

In this study, we propose a novel method to noninvasive BP estimation applying a combination of CNN and Transformer with the Parallel structure. By leveraging the strengths of both CNN and Transformer, we extract local and global features to enhance the feature representation using the channel and spatial attention mechanism. Our proposed model achieves Grade A of the BHS for BP predictions. It also meets the AAMI for BP predictions on MIMIC Dataset. Our results demonstrate the effectiveness of PCTN and its potential for noninvasive blood pressure estimation.

For future work, we plan to focus on developing lightweight models for estimating blood pressure that can be easily integrated into smart wearable devices [54]. Continuous monitoring of blood pressure using PPG signals will provide users with valuable insights into the daily fluctuations of their blood pressure, enabling them to make timely adjustments and maintain healthy blood pressure levels.

CRediT authorship contribution statement

Zhonghe Tian: Software, Data curation, Writing – original draft. **Aiping Liu:** Methodology, Writing – review & editing. **Guokang Zhu:** Investigation, Validation, Formal analysis. **Xun Chen:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors have no affiliation with any organization with a direct or indirect financial interest in the subject matter discussed in the manuscript.

Data availability

This study uses a publicly available dataset.

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